



Customer Shopping Behavior Analysis

Analysis of 3,900 transactions to uncover spending patterns, segments, product preferences, and subscription behavior to inform strategy.

Dataset Summary

Total Records

3,900 rows · 18 columns

Key Features

Demographics, purchase details,
shopping behavior, ratings,
shipping

Missing Data

37 missing values in Review Rating (imputed by category median)

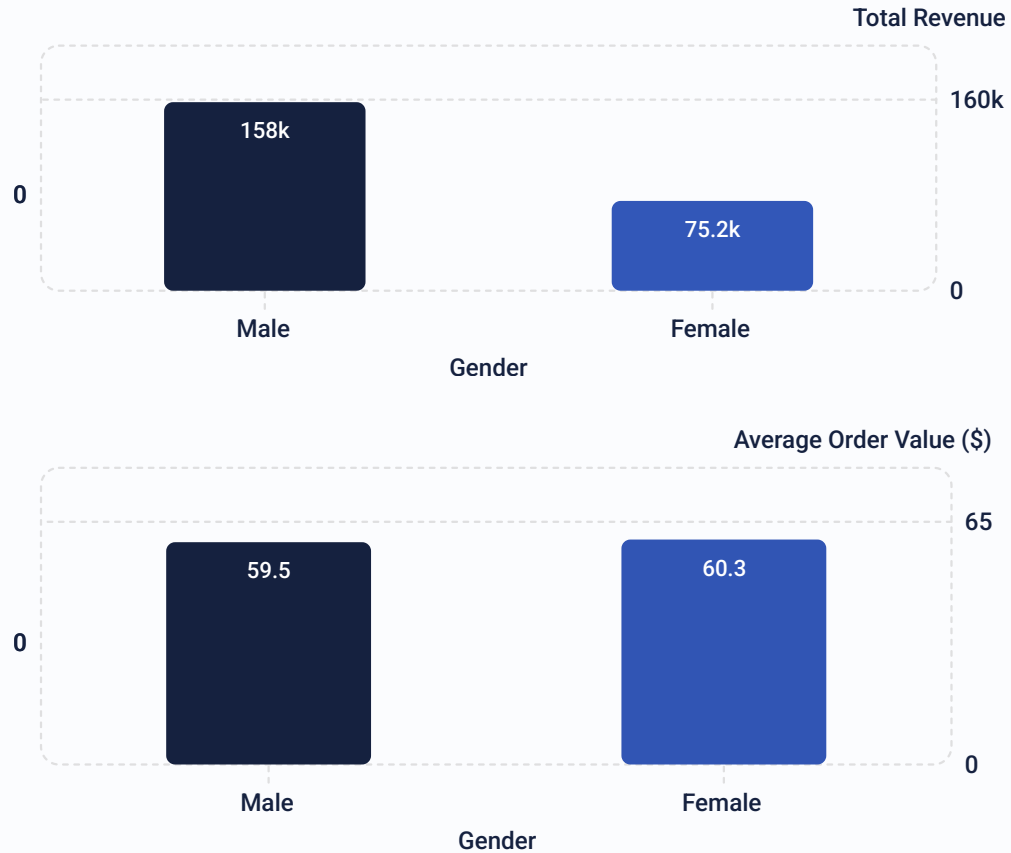


Data Preparation & Feature Engineering



- Loaded data with pandas; inspected with `df.head()`, `df.info()`, `df.describe()`
- Imputed missing ratings by product-category median
- Renamed columns to snake_case for clarity
- Engineered `age_group` and `purchase_frequency_days`
- Dropped redundant column `promo_code_used` (`discount_applied` retained)
- Loaded cleaned DataFrame to PostgreSQL for SQL queries

Revenue & Orders by Gender



Findings: Male customers purchase more than females while average order value is fairly the same across genders.



Discount Behavior & High-Spending Discount Users



Some of the products had half of its sale with discounted prices as these products rely heavily on discounts to drive sales we need to review our discount policies on these products to make sure that our margins do not vanish.

Actionable Insight

Review discount policy to balance volume and margins.

Top Rated vs Top Selling Products



Top rated products are not always top sellers, though top sellers still maintain fairly high ratings.

Top Sellers

Products with highest purchase counts per category.

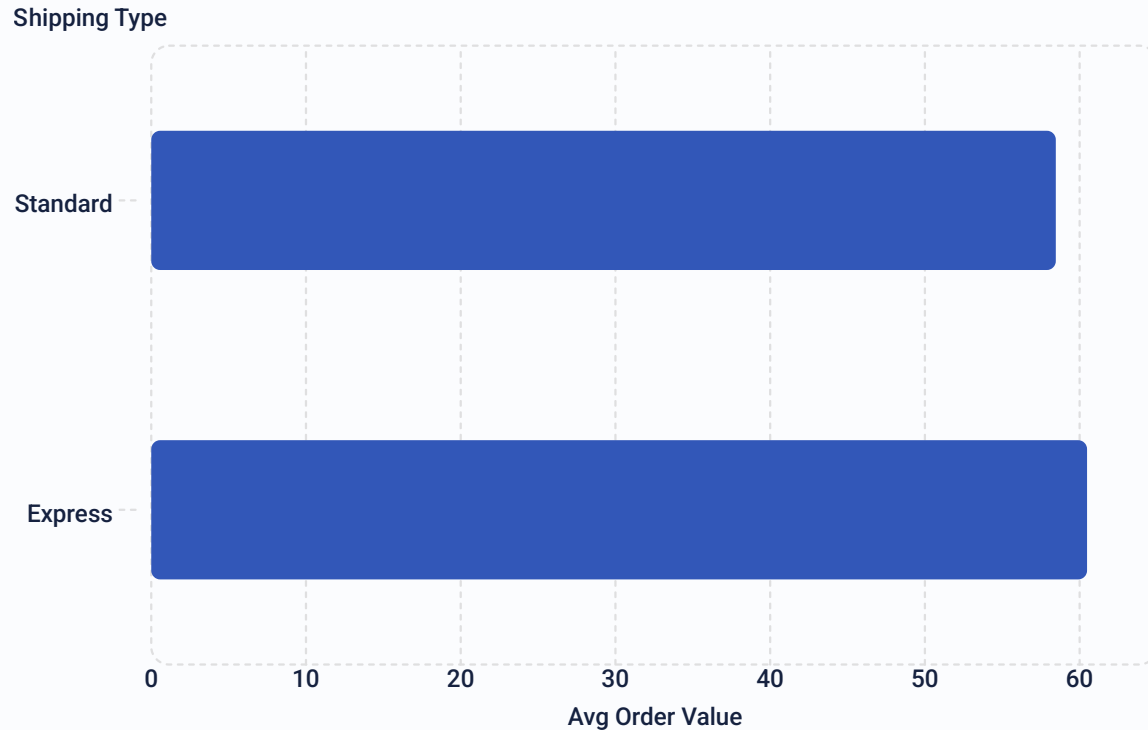
- Pants
- Jewellery
- Blouse

Top Rated

Products with highest average review ratings by category.

- Gloves
- Sandals
- Boots

Shipping & Subscription Insights



Findings: - Avg order value slightly higher for Express shipping. - Subscribers = 27% of customers and contribute ~26.88% of revenue; avg order value nearly the same as non-subscribers.



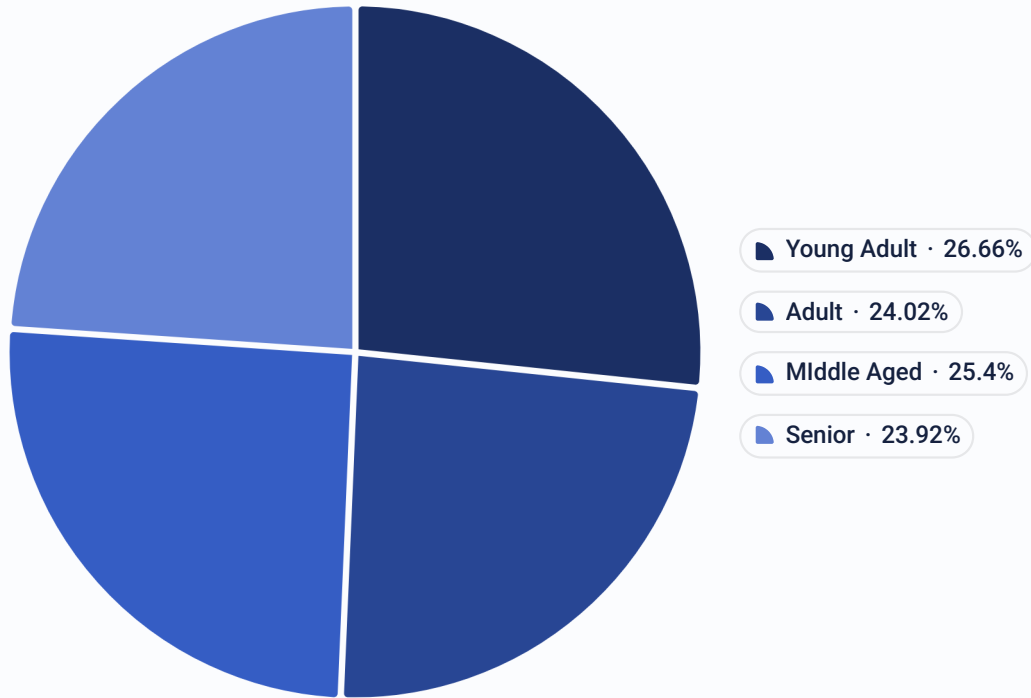
Customer Segmentation & Repeat Buyers



Segments: New, Returning, Loyal. Majority are loyal or returning; platform continues to attract new customers.

Repeat buyers (5+ purchases) are largely loyal, but only ~27.6% of loyal customers are subscribed.

Revenue by Age Group & Demographic Targets



Findings: Young adults contribute the most revenue; seniors the least. Difference between largest and smallest contribution is \$6,380 (Young Adults ~11.44% more than seniors).

Recommendation: Prioritize targeted marketing to Young Adults and initiatives to acquire more female customers.

Dashboard & Business Recommendations



Boost Subscriptions

Promote exclusive subscriber benefits to increase uptake.

Loyalty Programs

Reward repeat buyers to increase retention and CLV.

Review Discounts

Balance discount-driven volume with margin control.

Product Positioning & Shipping

Highlight top-rated and best-selling items; promote express shipping to lift AOV.